

The Strongest Teams for IKPC (strongestteams)

The IKPC¹ is right around the corner. The kindergarten teacher, Theresa, wants to form some teams for the competition. There are N children in her class, and she wants to form exactly K teams.



Figure 1: The forming of teams is a tedious task.

The children are sitting along a line, numbered from 0 to $N - 1$, according to their position in the line. Child i has a team play factor A_i and a programming skill B_i .

Each team must have at least one member, and no child can be assigned to multiple teams. Children $0 \leq i_1 < i_2 < \dots < i_l < N$ (for some $l > 0$) can form a team if for each $1 \leq p < l$ the following conditions hold:

- $A_{i_p} < A_{i_{p+1}}$, and
- there is no child with index $i_p < x < i_{p+1}$ such that $A_{i_p} < A_x$.

The strength of a team formed this way is $B_{i_1} + B_{i_2} + \dots + B_{i_l}$.

Your task is to find the maximum possible sum of team strengths that Theresa can achieve by forming exactly K teams!

Among the attachments of this task you may find a template file `strongestteams.*` with a sample incomplete implementation.

Input

The input file consists of:

- a line containing integers N, K .
- a line containing the N integers $A_0, \dots, A_{N-1}$.
- a line containing the N integers $B_0, \dots, B_{N-1}$.

¹International Kindergarten Programming Contest

Output

The output file must contain a single line consisting of 64-bit integer `ans`.

Constraints

- $1 \leq K \leq N \leq 100\,000$.
- $1 \leq A_i \leq N$ for each $i = 0 \dots N - 1$.
- $-10^9 \leq B_i \leq 10^9$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (11 points) $N \leq 20$.

- **Subtask 3** (9 points) $N \leq 5000$ and $B_i \geq 0$.

- **Subtask 4** (13 points) $B_i \geq 0$.

- **Subtask 5** (24 points) $N \leq 5000$.

- **Subtask 6** (43 points) No additional limitations.


Examples

input	output
4 1 1 2 2 3 10 -10 -2 3	10
5 2 5 2 4 3 2 4 3 3 4 3	10

Explanation

In the **first sample case** forming the team consisting only of the first child is optimal.

In the **second sample case** forming one team consisting of the first child and another team consisting of the second and third children is optimal.